

Binomial Supply-Chain Depth Model

Raw Material Access · Processing Steps · Logistics Distance
Plus-Sum Labour Dynamics · Consumer Price Formation

An extension of the binomial commodity wave framework
that traces value accumulation and cost compounding
across each processing stage in a supply chain,
adjusted for labour's plus-sum contribution, logistic distance,
and supply-chain depth d —from single-stage fresh produce
to intermediate concrete, to deeply nested
complex manufactured goods such as automobiles and smartphones.

Contents

1	Motivation and Scope	2
2	Foundations: Single-Stage Binomial Revisited	2
3	Multi-Stage Supply-Chain Model	3
3.1	Stage Definition and Notation	3
3.2	Stage-by-Stage Cost Propagation	3
3.3	Recursive Expansion	4
3.4	Plus-Sum Labour Contribution Across the Chain	4
3.5	Logistic Distance and Global vs. Local Chains	5
4	Canonical Product Classes	5
4.1	Class I — Fresh Produce ($d = 1-2$)	5
4.1.1	Model	5
4.1.2	Characteristics	5
4.2	Class II — Concrete ($d = 4-7$)	6
4.2.1	Supply-chain decomposition	6
4.2.2	Model	6
4.2.3	Key structural features	6
4.3	Class III — Complex Manufactured Goods ($d = 8-25+$)	7
4.3.1	Parallel sub-chain architecture	7
4.3.2	Smartphone sub-chains	7
4.3.3	Rare-earth bottleneck: binomial scarcity at high depth	7
5	Plus-Sum Labour Dynamics Across Chain Depth	8
5.1	Labour Share as a Function of Depth	8
5.2	Wage Gradient and Skill Composition	8
5.3	Automation Displacement Within the Chain	9
6	Disruption Propagation and Cascade Dynamics	9
6.1	Single-Stage Disruption	9
6.2	Correlated Disruption Across Sub-chains	9
7	Cross-Class Comparison and Unified Summary Table	10
7.1	Analytical comparison across product classes	10
7.2	Consumer price decomposition identity	10
8	Parameter Reference	11
	Concluding Remarks	11

§1. Motivation and Scope

The commodity pricing model established in Model I treats a raw material as a single extractable quantity whose price is governed by a binomial access process and an inverse-exponential abundance decay. Real goods reaching consumers are not single-stage commodities—they are the output of a *chain of transformations*, each of which:

1. introduces a new binomial access trial (can this processing stage obtain its required inputs?);
2. adds labour at that stage (a plus-sum contributor);
3. adds logistic cost proportional to distance travelled;
4. amplifies or damps the upstream price signal depending on stage-specific margins and substitution possibilities.

We formalise this as the *supply-chain depth model*, parameterised by $d \in \mathbb{Z}_{>0}$, the number of distinct processing stages between raw earth and final consumer. Three canonical product classes anchor the framework:

Product class	d	Raw material inputs	Chains	Example
Fresh produce	1–2	Soil, water, seed	1	Tomatoes, wheat grain
Processed agriculture	2–4	Grain, energy, water	2–3	Flour, cooking oil
Construction material	4–7	Limestone, clay, fuel	3–5	Concrete, steel rebar
Complex manufactured	8–15	Ore, silicon, hydrocarbons	6–12	Automobile
Advanced technology	12–25+	Rare earths, speciality gas	10–20	Smartphone

§2. Foundations: Single-Stage Binomial Revisited

From the master framework, the cost of obtaining one unit of a raw material after n binomial access trials with failure probability p is:

$$C_0(n, p) = c_0 (1 - p)^{-n}, \quad (1)$$

where c_0 is the baseline extraction cost and $(1 - p)^{-n}$ is the scarcity multiplier. In the post-access abundance regime:

$$C_0(t) = C_{\text{peak}} e^{-\lambda t} + C_{\text{floor}}, \quad C_{\text{floor}} = \frac{\alpha_0}{I \cdot \rho}, \quad (2)$$

with infrastructure quality I and material density ρ .

This single-stage cost C_0 becomes the *input cost* to stage 1 of the supply chain. Every subsequent stage transforms it, adds value, incurs logistics cost, and passes the result forward.

§3. Multi-Stage Supply-Chain Model

3.1. Stage Definition and Notation

Definition 3.1 (Supply-chain stage). *Stage $k \in \{1, \dots, d\}$ is a discrete processing step that takes as input a bundle of intermediate goods and produces a single intermediate output. Each stage is characterised by:*

- p_k — binomial failure probability (can the stage obtain inputs?);
- ℓ_k — labour hours per unit output at stage k ;
- w_k — wage rate at stage k (plus-sum: includes skill premium);
- d_k — logistic distance (km) between stage $k-1$ and stage k ;
- τ_k — transport cost per unit per km at stage k ;
- m_k — material conversion ratio (units input per unit output);
- η_k — energy intensity (kWh per unit processed);
- e_k — energy price at stage k .

Definition 3.2 (Supply-chain depth d). *The depth d of a product is the minimum number of stages required to transform raw earth materials into a unit of the final consumer good. Depth is additive across parallel sub-chains when they are assembled:*

$$d_{\text{assembly}} = \max_j (d_j) + 1,$$

where the $\max$ captures the critical-path sub-chain (the bottleneck) and $+1$ accounts for the final assembly stage.

3.2. Stage-by-Stage Cost Propagation

Let C_k denote the *unit cost at the output of stage k* (i.e., the cost passed forward to stage $k+1$ as its input cost). Define $C_0 \equiv C_0(n, p)$ from (1)–(2) as the raw material extraction cost.

For each stage $k = 1, \dots, d$:

$$C_k = \underbrace{m_k C_{k-1} \cdot (1 - p_k)^{-n_k}}_{\text{input cost + scarcity}} + \underbrace{\ell_k w_k}_{\text{labour (plus-sum)}} + \underbrace{\eta_k e_k}_{\text{energy}} + \underbrace{\tau_k d_k}_{\text{logistics}} + \underbrace{\mu_k C_{k-1}}_{\text{stage margin}}, \quad (3)$$

where:

- $m_k \geq 1$ is the material input ratio (waste, spoilage, yield loss— $m_k = 1$ for lossless transformation);

- $(1 - p_k)^{-n_k}$ is the binomial scarcity amplifier at stage k (high if stage k 's inputs are rare or fragile);
- $\ell_k w_k$ is the plus-sum labour contribution (labour adds new value; this term is *not* a transfer from an existing pool);
- $\mu_k \geq 0$ is the stage markup.

The final consumer price is:

$$P_{\text{fin}} = C_d \cdot (1 + \mu_{\text{retail}}), \quad (4)$$

with retail margin μ_{retail} applied to the accumulated cost stack C_d .

3.3. Recursive Expansion

Unrolling (3) recursively:

$$P_{\text{fin}} = (1 + \mu_{\text{ret}}) \left[\underbrace{\prod_{k=1}^d (m_k(1 - p_k)^{-n_k} + \mu_k)}_{\text{scarcity-margin compounding}} \cdot C_0 + \sum_{k=1}^d \underbrace{\prod_{j=k+1}^d (m_j(1 - p_j)^{-n_j} + \mu_j)}_{\text{downstream amplification of stage-}k \text{ cost}} \cdot (\ell_k w_k + \eta_k e_k + \tau_k d_k) \right]. \quad (5)$$

Remark 3.1 (Downstream amplification of upstream labour). *Each labour, energy, and logistic cost incurred at stage k is amplified by the product of all downstream scarcity-and-margin factors $\prod_{j>k} (m_j(1 - p_j)^{-n_j} + \mu_j)$. A disruption or wage increase at an early processing stage propagates to the consumer price with compounding force—a phenomenon empirically observed in semiconductor supply-chain shocks (2020–2022) and fertiliser price cascades (2021–2022).*

3.4. Plus-Sum Labour Contribution Across the Chain

Define the *total plus-sum value added* by labour across all stages:

$$\mathcal{V}_L = \sum_{k=1}^d \ell_k w_k \cdot \prod_{j=k+1}^d (m_j(1 - p_j)^{-n_j} + \mu_j), \quad (6)$$

and the *raw-material pass-through* (the fraction of final price attributable to earth inputs):

$$\phi_{\text{raw}} = \frac{\prod_{k=1}^d (m_k(1 - p_k)^{-n_k} + \mu_k) \cdot C_0}{P_{\text{fin}} / (1 + \mu_{\text{ret}})}. \quad (7)$$

For simple products (d small, p_k low) ϕ_{raw} is large; for complex products (d large, specialised labour dominant) ϕ_{raw} is small and $\mathcal{V}_L / P_{\text{fin}}$ dominates.

3.5. Logistic Distance and Global vs. Local Chains

The logistic term $\tau_k d_k$ is linear in distance but enters the recursive formula amplified by downstream factors. Define the *logistic depth penalty*:

$$L_{\text{fin}} = \sum_{k=1}^d \tau_k d_k \cdot \prod_{j=k+1}^d (m_j(1-p_j)^{-n_j} + \mu_j). \quad (8)$$

For global chains where d_k is large (ocean freight, transcontinental rail), L_{fin} is significant but often smaller than labour and energy terms. Its importance rises sharply when:

- transport cost τ_k spikes (fuel price shock, port congestion);
- the disrupted stage is early (k small), maximising downstream amplification;
- the product is high-volume, low-value (fresh produce, construction aggregate) where logistics is a large share of P_{fin} .

§4. Canonical Product Classes

4.1. Class I — Fresh Produce ($d = 1-2$)

4.1.1. Model

Fresh fruits and vegetables undergo at most two stages: agricultural production (stage 1) and cold-chain distribution (stage 2).

$$\begin{aligned} C_1 &= m_1 C_0 (1 - p_1)^{-n_1} + \ell_1 w_1 + \eta_1 e_1 \\ C_2 &= m_2 C_1 (1 - p_2)^{-n_2} + \tau_2 d_2 \\ P_{\text{fin}} &= C_2 (1 + \mu_{\text{ret}}) \end{aligned}$$

4.1.2. Characteristics

- **Low scarcity compounding:** $(1 - p_1)^{-n_1}$ and $(1 - p_2)^{-n_2}$ are close to 1 under normal conditions (soil and water accessible, weather reliable).
- **Labour is the dominant plus-sum term:** harvesting, packing, and handling account for 30–60% of shelf price for labour-intensive crops (strawberries, lettuce).
- **Perishability imposes a hard logistic constraint:** d_2 must satisfy $d_2/v_{\text{transport}} < T_{\text{shelf}}$, where T_{shelf} is the product's perishable lifetime. This caps the feasible supply radius and creates a *natural local-market advantage* not present in durable goods.
- **Drought / frost shock:** a bad-weather event drives $p_1 \rightarrow p_1 + \Delta p$ temporarily, spiking C_1 via the binomial amplifier even though C_0 (soil cost) is unchanged.

Example 4.1 (Tomatoes: local vs. imported). Let $c_0 = 0.05$ \$/kg, $p_1 = 0.08$, $n_1 = 1$ (single growing season trial), $\ell_1 w_1 = 0.30$ \$/kg, $m_1 = 1.3$ (30% field waste):

$$C_1 = 1.3 \times 0.05 \times (0.92)^{-1} + 0.30 = 0.071 + 0.30 = 0.371 \text{ \$/kg.}$$

Local distribution ($d_2 = 80 \text{ km}$, $\tau_2 = 0.0008 \text{ \$/kg km}$): $P_{\text{fin}}^{\text{local}} = (0.371 + 0.064) \times 1.35 \approx 0.59 \text{ \$/kg}$.

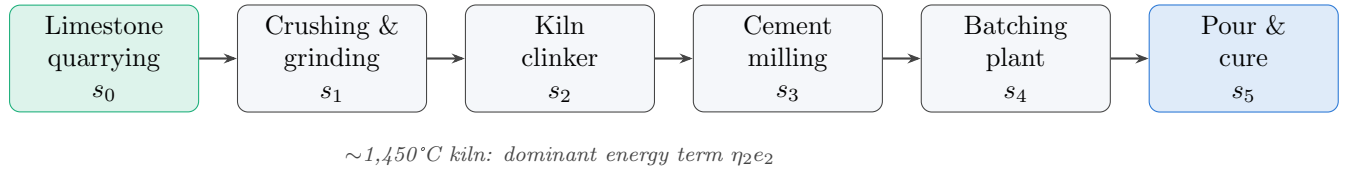
Global import ($d_2 = 4,000 \text{ km}$): $P_{\text{fin}}^{\text{global}} = (0.371 + 3.20) \times 1.35 \approx 4.82 \text{ \$/kg}$.

The 50× distance increase multiplies final price by 8× — demonstrating that for low-value, high-perishability goods the logistic term dominates even though it is structurally “early” in the chain.

4.2. Class II — Concrete ($d = 4\text{--}7$)

4.2.1. Supply-chain decomposition

Portland cement concrete involves five primary stages:



4.2.2. Model

$$C_1 = m_1 C_0 (1 - p_1)^{-n_1} + \ell_1 w_1 + \eta_1 e_1 + \tau_1 d_1 \quad (\text{quarry to crusher}), \quad (9)$$

$$C_2 = m_2 C_1 (1 - p_2)^{-n_2} + \ell_2 w_2 + \eta_2 e_2 \quad (\text{kiln; dominant energy term: } \eta_2 e_2 \gg \ell_2 w_2), \quad (10)$$

$$C_3 = m_3 C_2 + \ell_3 w_3 + \eta_3 e_3 \quad (\text{milling; } p_3 \approx 0, \text{ no scarcity}), \quad (11)$$

$$C_4 = m_4 (C_3 + C_{\text{agg}} + C_{\text{water}}) + \ell_4 w_4 + \tau_4 d_4 \quad (\text{batching with aggregate and water}), \quad (12)$$

$$C_5 = C_4 + \ell_5 w_5 + \tau_5 d_5 \quad (\text{on-site pour and cure}). \quad (13)$$

4.2.3. Key structural features

- **Energy intensity at stage 2** dominates: clinker production consumes $\sim 3.4 \text{ GJ/tonne}$, making $\eta_2 e_2$ the largest single cost term. Energy price shocks propagate forward amplified by $m_3 m_4 m_5 (1 + \mu_{\text{ret}})$.
- **Multiple parallel raw material chains:** aggregate (sand, gravel) and water enter at stage 4 as separate binomial processes, introducing *parallel chain depth*:

$$d_{\text{concrete}} = \max(d_{\text{cement}}, d_{\text{aggregate}}, d_{\text{water}}) + 1.$$

- **Low labour share at consumer price:** despite multiple stages, concrete is capital- and energy-intensive; $\mathcal{V}_L / P_{\text{fin}} \approx 15\text{--}25\%$ against $50\text{--}70\%$ for fresh produce.
- **Logistics is local by necessity:** concrete is non-perishable but *hardens within ~ 90 minutes* of batching, imposing the same effective radius constraint as fresh produce via a chemical (not biological) perishability.

4.3. Class III — Complex Manufactured Goods ($d = 8\text{--}25+$)

4.3.1. Parallel sub-chain architecture

A smartphone or automobile is not a single linear chain of depth d but a *directed acyclic graph* (DAG) of parallel sub-chains that converge at assembly stages. Let $\mathcal{T}$ be the set of sub-chains indexed by i , each of depth d_i . The final assembly stage adds d_{asm} additional stages.

Definition 4.1 (DAG product cost).

$$C_{\text{asm}} = \sum_{i \in \mathcal{T}} q_i C_{d_i}^{(i)} + \ell_{\text{asm}} w_{\text{asm}} + \tau_{\text{asm}} d_{\text{asm}}, \quad (14)$$

where q_i is the number of units of sub-chain i 's output required per finished product (e.g. $q_{\text{battery cell}} = 6,000$ for an EV battery pack), and $C_{d_i}^{(i)}$ is the per-unit cost of sub-chain i 's final stage.

4.3.2. Smartphone sub-chains

Sub-chain i	Key raw material	d_i	q_i
Silicon wafer / SoC	Quartz sand, dopant gases	12–18	1
Display (OLED)	Indium, rare earths, polymer	10–14	1
Battery (Li-ion)	Lithium, cobalt, graphite	7–10	1
Camera module	Rare earth magnets, borosilicate	8–12	3–5
Structural frame	Aluminium, titanium alloy	5–8	1
PCB / passive components	Copper, tin, tantalum	6–10	100+
Software / firmware	Human intellectual labour only	0	1
Final assembly	—	1–2	1

Remark 4.1 (Software as a pure plus-sum, zero-material stage). *The software sub-chain has $d_i = 0$: no raw material extraction trial, no conversion ratio $m_k > 1$, no logistics cost. Its contribution to C_{asm} is pure labour:*

$$C_{\text{sw}} = \ell_{\text{sw}} w_{\text{sw}},$$

with w_{sw} reflecting skill-intensive wages. This is the limit of plus-sum labour: value is created from human cognition alone, with $\phi_{\text{raw}} = 0$ for this sub-chain. It also means software is infinitely scalable ($q_i = \infty$ at zero marginal cost post-development)—the one component of a complex product that escapes the binomial scarcity constraint entirely.

4.3.3. Rare-earth bottleneck: binomial scarcity at high depth

Critical minerals in Class III products (cobalt, lithium, indium, neodymium) often combine high depth with high failure probability p_k . The scarcity amplifier at stage k in a deep chain compounds all downstream stages:

$$\frac{\partial P_{\text{fin}}}{\partial p_k} = m_k n_k (1 - p_k)^{-n_k-1} C_{k-1} \cdot \prod_{j=k+1}^d (m_j (1 - p_j)^{-n_j} + \mu_j) \cdot (1 + \mu_{\text{ret}}). \quad (15)$$

The partial derivative (15) grows with both the depth of the rare-earth stage k (via the downstream product) and the current scarcity level (via $(1 - p_k)^{-n_k-1}$). This means small increases in cobalt or lithium mining difficulty propagate to consumer prices with *superlinear* force in deep chains.

§5. Plus-Sum Labour Dynamics Across Chain Depth

5.1. Labour Share as a Function of Depth

Define the *labour share of final consumer price*:

$$\Lambda(d) = \frac{\mathcal{V}_L}{P_{\text{fin}}/(1 + \mu_{\text{ret}})} = \frac{\sum_{k=1}^d \ell_k w_k \prod_{j=k+1}^d (m_j (1 - p_j)^{-n_j} + \mu_j)}{C_d}. \quad (16)$$

Theorem 5.1 (Labour share monotonicity under mild conditions). *If all stages have $p_k < 1/2$ and $m_k \leq 2$, and if wage rates w_k grow with stage specialisation (i.e. w_k is non-decreasing in k), then $\Lambda(d)$ is non-decreasing in d for fixed C_0 .*

Sketch. Adding a stage $d+1$ increases the numerator by $\ell_{d+1} w_{d+1}$ (a strictly positive term) and multiplies all prior numerator terms by $(m_{d+1}(1 - p_{d+1})^{-n_{d+1}} + \mu_{d+1}) > 1$. The denominator is the new C_{d+1} , which also increases, but by the same multiplicative factor applied to the prior denominator *plus* the new labour term. Provided $\ell_{d+1} w_{d+1}$ is positive, the ratio does not decrease. $\square$

Remark 5.1. *This theorem formalises the intuition that complex manufactured goods are labour-value goods first and material-cost goods second. A smartphone costing \$1,200 has perhaps \$40 of raw earth material content; the rest is accumulated skill and capital across deep processing chains.*

5.2. Wage Gradient and Skill Composition

Across a chain of depth d , assume wage rates follow:

$$w_k = w_1 e^{\sigma(k-1)/(d-1)}, \quad \sigma \geq 0, \quad (17)$$

where σ is the *skill premium exponent*. For $\sigma = 0$, all stages pay the same wage (undifferentiated labour). For $\sigma > 0$, late-stage (high-skill) workers earn exponentially more than early-stage (raw extraction) workers—matching the empirical wage distribution in manufacturing and technology industries.

The total labour value added is:

$$\mathcal{V}_L = w_1 \sum_{k=1}^d \ell_k e^{\sigma(k-1)/(d-1)} \cdot \prod_{j=k+1}^d (m_j + \mu_j). \quad (18)$$

5.3. Automation Displacement Within the Chain

Automation at stage k reduces ℓ_k toward zero but raises η_k (energy for machines) and m_k is often reduced (better precision). Define automation penetration $a_k \in [0, 1]$:

$$\ell_k^{\text{eff}} = \ell_k(1 - a_k), \quad \eta_k^{\text{eff}} = \eta_k(1 + \beta_{\text{mach}} a_k), \quad m_k^{\text{eff}} = m_k(1 - \gamma_{\text{yield}} a_k), \quad (19)$$

where $\beta_{\text{mach}} > 0$ is the machine energy overhead and $\gamma_{\text{yield}} \in [0, 1)$ is the yield improvement from automation. The net effect on stage cost depends on whether energy savings and yield improvement offset the displaced-labour saving—a key empirical question in manufacturing economics.

§6. Disruption Propagation and Cascade Dynamics

6.1. Single-Stage Disruption

A disruption at stage k is modelled as a temporary increase in failure probability: $p_k \rightarrow p_k + \Delta p_k$ for T_{dis} periods. The resulting spike in final consumer price is:

$$\Delta P_{\text{fin}} \approx \frac{\partial P_{\text{fin}}}{\partial p_k} \cdot \Delta p_k = m_k n_k (1 - p_k)^{-n_k - 1} C_{k-1} \cdot \prod_{j>k} (\cdot) \cdot \Delta p_k \cdot (1 + \mu_{\text{ret}}). \quad (20)$$

The downstream amplification product $\prod_{j>k}$ means:

- Early-stage disruptions (small k) propagate through the most downstream stages and produce the largest consumer price spikes.
- Late-stage disruptions (large k) are amplified by fewer stages but may affect time-critical assembly processes (just-in-time manufacturing).

6.2. Correlated Disruption Across Sub-chains

In a DAG product (14) with $|\mathcal{T}|$ sub-chains, a shared input (e.g. energy, speciality gas, port access) may disrupt multiple sub-chains simultaneously. Let disruption indicator $\xi_i \in \{0, 1\}$ for sub-chain i . If disruptions are correlated with correlation ρ_{ij} :

$$\text{Var} \left[\sum_i q_i \Delta C^{(i)} \right] = \sum_i q_i^2 \text{Var}[\Delta C^{(i)}] + 2 \sum_{i<j} q_i q_j \rho_{ij} \sqrt{\text{Var}[\Delta C^{(i)}] \text{Var}[\Delta C^{(j)}]}. \quad (21)$$

For positively correlated sub-chains ($\rho_{ij} > 0$, e.g. all sub-chains sourcing from the same geography), the variance of the final product cost shock is *superadditive*, explaining why global supply-chain concentration creates disproportionate price volatility.

§7. Cross-Class Comparison and Unified Summary Table

7.1. Analytical comparison across product classes

Property	Fresh duce	pro- Concrete	Automobile	Smartphone
Depth d	1–2	4–7	8–12	12–25+
Sub-chains $ \mathcal{T} $	1	3–4	50–200	200–1,000+
ϕ_{raw} (material share)	15–30%	25–40%	8–15%	3–8%
Λ (labour share)	40–60%	15–25%	30–45%	20–40%
Dominant cost term	$\ell_k w_k, \tau d$	ηe (kiln)	Assembly ℓw	R&D + IP ℓw
Logistic sensitivity	Very high (perishable)	High (chemi- cal)	Medium	Low (air freight)
Scarcity amplifier	Weather (Δp volatile)	Limestone (low p)	Steel / chips	Rare earths (high p)
Automation a_k effect	Moderate	High at $k = 2, 3$	High at as- sembly	Very high at fab

7.2. Consumer price decomposition identity

For any product of depth d , the final price satisfies the *supply-chain value identity*:

$$P_{\text{fin}} = (1 + \mu_{\text{ret}}) \left[\underbrace{\Phi_{\text{raw}} \cdot C_0}_{\text{material origin}} + \underbrace{\mathcal{V}_L}_{\text{plus-sum labour}} + \underbrace{L_{\text{fin}}}_{\text{logistics}} + \underbrace{\mathcal{E}}_{\text{energy stack}} + \underbrace{\mathcal{M}}_{\text{margin stack}} \right], \quad (22)$$

where $\Phi_{\text{raw}} = \prod_{k=1}^d (m_k(1 - p_k)^{-n_k} + \mu_k)$ is the scarcity-compounding factor applied to the raw material cost.

Identity (22) unifies the single-stage commodity model with the multi-stage supply chain: when $d = 1$ and $\ell_1 = \tau_1 = 0$, it reduces exactly to (1). The plus-sum nature of labour $\mathcal{V}_L > 0$ is the fundamental reason deep chains create more total value than shallow ones—labour at each stage adds to the pie rather than merely redistributing an existing pool, in direct contrast to the zero-sum casino model where $\mathcal{V}_L = 0$ by construction.

§8. Parameter Reference

Symbol	Domain	Definition	Typical range
d	$\mathbb{Z}_{>0}$	Supply-chain depth (stages)	1–25+
p_k	$(0, 1)$	Stage- k binomial failure probability	$[0.01, 0.6]$
n_k	$\mathbb{Z}_{>0}$	Trials per stage	1–50
m_k	$[1, \infty)$	Material conversion ratio (input/output)	1.0–3.0
ℓ_k	$\mathbb{R}_{\geq 0}$	Labour hours per unit at stage k	0–100
w_k	$\mathbb{R}_{>0}$	Wage rate at stage k (\$/hr)	2–500
σ	$\mathbb{R}_{\geq 0}$	Skill premium exponent	0–3
η_k	$\mathbb{R}_{\geq 0}$	Energy intensity (kWh/unit)	0–10,000
e_k	$\mathbb{R}_{>0}$	Energy price at stage k (\$/kWh)	0.05–0.50
d_k	$\mathbb{R}_{\geq 0}$	Logistic distance (km)	0–20,000
τ_k	$\mathbb{R}_{\geq 0}$	Transport cost (\$/unit/km)	10^{-5} – 10^{-2}
μ_k	$\mathbb{R}_{\geq 0}$	Stage markup (fraction of input cost)	0.02–0.40
μ_{ret}	$\mathbb{R}_{\geq 0}$	Retail margin	0.10–0.60
a_k	$[0, 1]$	Automation penetration at stage k	0–0.95
β_{mach}	$\mathbb{R}_{>0}$	Machine energy overhead factor	0.1–1.0
ϕ_{raw}	$(0, 1)$	Raw material share of final price	0.03–0.50
$\Lambda(d)$	$(0, 1)$	Labour share of final price	0.10–0.70
Φ_{raw}	$[1, \infty)$	Scarcity-compounding factor	1 – 10^6
$\mathcal{V}_L$	$\mathbb{R}_{\geq 0}$	Total plus-sum labour value added	—
L_{fin}	$\mathbb{R}_{\geq 0}$	Total logistics cost (amplified)	—

Concluding Remarks

The supply-chain depth model makes two structural claims that generalise the original binomial commodity framework.

First, scarcity compounds across stages. A disruption to an upstream raw material does not simply raise that material’s price in isolation—it is amplified by every downstream conversion ratio, margin, and scarcity factor between the point of disruption and the consumer. This compounding is superlinear in the depth d , explaining why rare earth shortages in the mining sector (k small) produce semiconductor shortages in finished electronics (k large) within months.

Second, labour is the countervailing plus-sum force. At every stage, $\ell_k w_k$ enters the cost stack not as a redistribution of existing value but as new value created by human time

and skill applied to capital. Deep chains (d large) therefore contain more accumulated plus-sum value than shallow chains—a fact obscured by the visibility of raw material prices (which are quoted on commodity exchanges) versus labour’s diffuse contribution across many invisible processing stages.

The interplay between these two forces—scarcity compounding Φ_{raw} amplifying upstream shocks, and labour value $\mathcal{V}_L$ growing with chain depth—determines the resilience and price structure of every manufactured good. Products high in both (rare-earth electronics) are simultaneously the most valuable and the most vulnerable to upstream disruption. Products high in labour but low in scarcity (fresh produce under favourable conditions) are vulnerable instead through the logistic channel L_{fin} , where distance and perishability are the binding constraints. Products high in energy and capital but low in labour (concrete, bulk steel) depend on the stability of energy prices e_k , which themselves follow a commodity price curve with their own binomial access wave.

In each case, the framework reduces to the same master equation: binomial access governs what flows in, plus-sum labour governs what value is added, and logistic distance governs how much of the surplus is consumed in transit before the consumer takes delivery.